

Named Entity Recognition (NER)

(Natural Language Processing)

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Named Entity Recognition (NER)

- **Find** and **classify** all the named entities in a text.
- What is a named entity?
 - A mention of an entity using its name.
 - Kansas Jayhawks
 - This is a subset of the possible mentions...
 - Kansas, Jayhawks, the team, it, they
- **Find** means identify the exact span of the mention
- **Classify** means determine the category of the entity being referred to

Proper Nouns	Common Nouns
Havaianas	slippers
Coconut	tree
Jolibee	fastfood
Acer	computer
Robert	man/boy
Maria	woman/girl

Example

I hear Place **Berlin** is wonderful in the Time **winter**

- The category can be generic like **Organization, Person, Location, Time,** etc., or a custom category depending on the use case such as **Healthcare Terms, Programming Language,** etc.
- For example, an NER model detects “football” as an entity in a paragraph and classifies it into the category of sports.

Example

- The decision by the independent MP **Andrew Wilkie** to withdraw his support for the minority **Labor** government sounded dramatic but it should not further threaten its stability. When, after the **2010** election, **Wilkie**, **Rob Oakeshott**, **Tony Windsor** and the **Greens** agreed to support **Labor**, they gave just two guarantees: confidence and supply.

Example

- The decision by the independent MP **Andrew Wilkie** to withdraw his support for the minority **Labor** government sounded dramatic but it should not further threaten its stability. When, after the **2010** election, **Wilkie**, **Rob Oakeshott**, **Tony Windsor** and the **Greens** agreed to support **Labor**, they gave just two guarantees: confidence and supply.

Person, **Date**, **Organization**

NE Types

Type	Tag	Sample Categories
People	PER	Individuals, fictional characters, small groups
Organization	ORG	Companies, agencies, political parties, religious groups, sports teams
Location	LOC	Physical extents, mountains, lakes, seas
Geo-Political Entity	GPE	Countries, states, provinces, counties
Facility	FAC	Bridges, buildings, airports
Vehicles	VEH	Planes, trains, and automobiles

Type	Example
People	<i>Turing</i> is often considered to be the father of modern computer science.
Organization	The <i>IPCC</i> said it is likely that future tropical cyclones will become more intense.
Location	The <i>Mt. Sanitas</i> loop hike begins at the base of <i>Sunshine Canyon</i> .
Geo-Political Entity	<i>Palo Alto</i> is looking at raising the fees for parking in the University Avenue district.
Facility	Drivers were advised to consider either the <i>Tappan Zee Bridge</i> or the <i>Lincoln Tunnel</i> .
Vehicles	The updated <i>Mini Cooper</i> retains its charm and agility.

If your task is to find out
‘where’,
‘what’,
‘who’,
‘when’ from a sentence,

NER is the solution.

Ambiguity in NE

Name	Possible Categories
<i>Washington</i>	Person, Location, Political Entity, Organization, Facility
<i>Downing St.</i>	Location, Organization
<i>IRA</i>	Person, Organization, Monetary Instrument
<i>Louis Vuitton</i>	Person, Organization, Commercial Product

[*PERS* Washington] was born into slavery on the farm of James Burroughs.

[*ORG* Washington] went up 2 games to 1 in the four-game series.

Blair arrived in [*LOC* Washington] for what may well be his last state visit.

In June, [*GPE* Washington] passed a primary seatbelt law.

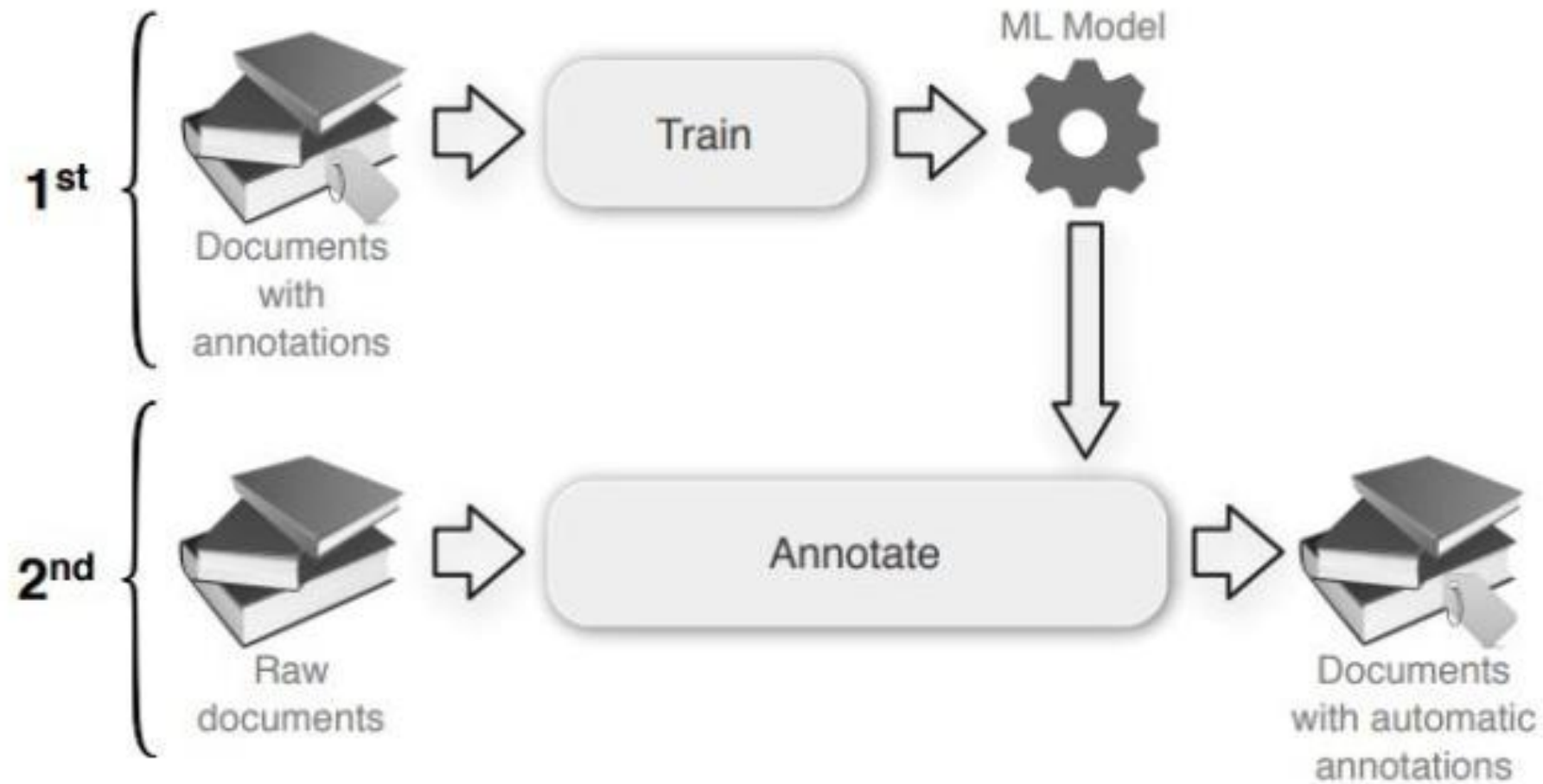
The [*FAC* Washington] had proved to be a leaky ship, every passage I made...

[Named Entity Visualizer](#)

NER Approaches

- As with partial parsing and chunking there are two basic approaches (and hybrids)
 - Rule-based (regular expressions)
 - Lists of names
 - Patterns to match things that look like names
 - Patterns to match the environments that classes of names tend to occur in.
 - ML-based approaches
 - Get annotated training data
 - Extract features
 - Train systems to replicate the annotation

NER Pipeline



NER Features

- Features may include
 - the word
 - POS tag
 - IOB tag
 - Inside
 - Outside
 - Beginning
 - the shape of the word

Features					Label
American	NNP	B _{NP}	cap		B _{ORG}
Airlines	NNPS	I _{NP}	cap		I _{ORG}
,	PUNC	O	punc		O
a	DT	B _{NP}	lower		O
unit	NN	I _{NP}	lower		O
of	IN	B _{PP}	lower		O
AMR	NNP	B _{NP}	upper		B _{ORG}
Corp.	NNP	I _{NP}	cap_punc		I _{ORG}
,	PUNC	O	punc		O
immediately	RB	B _{ADVP}	lower		O
matched	VBD	B _{VP}	lower		O
the	DT	B _{NP}	lower		O
move	NN	I _{NP}	lower		O
,	PUNC	O	punc		O
spokesman	NN	B _{NP}	lower		O
Tim	NNP	I _{NP}	cap		B _{PER}
Wagner	NNP	I _{NP}	cap		I _{PER}
said	VBD	B _{VP}	lower		O
.	PUNC	O	punc		O

Evaluation of NER

- The 2-by-2 contingency table

	Correct	Not correct
Selected	True-Positive	False-Positive
Not Selected	False-Negative	True-Negative

Evaluation of NER

- The 2-by-2 contingency table

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Selected and Not selected (Predicted Values)

Correct and not correct (Actual Values)

Evaluation of NER

	Correct	Not correct
Selected	True-Positive	False-Positive
Not Selected	False-Negative	True-Negative

Precision: % of selected items that are correct = $tp / (tp + fp)$

- From all the classes we have predicted as positive, how many are actually positive.
- Precision should be high as possible.

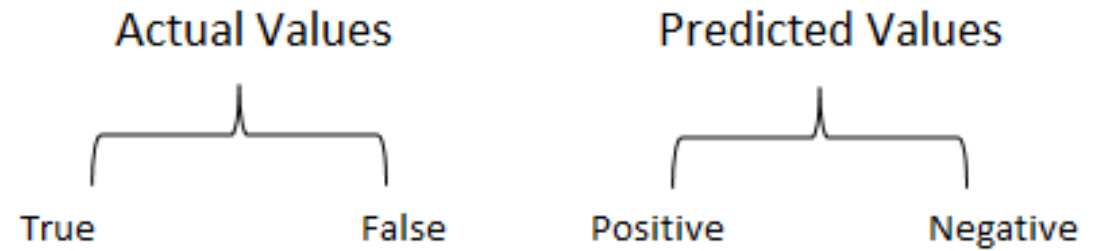
Evaluation of NER

	Correct	Not correct
Selected	True-Positive	False-Positive
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Recall: % of correct items that are selected = $tp / (tp + fn)$

- From all the positive classes, how many we predicted correctly.
- Recall should be high as possible.

Precision & Recall



y	y pred	output for threshold 0.6	Recall	Precision	Accuracy
0	0.5	0			
1	0.9	1			
0	0.7	1			
1	0.7	1			
1	0.3	0			
0	0.4	0			
1	0.5	0			

$$\textbf{Precision} = \frac{TP}{TP + FP}$$

$$\textbf{Recall} = \frac{TP}{TP + FN}$$

$$\textbf{Accuracy} = TP + TN / \textbf{Total}$$

Evaluation of NER

- A combined measure: F

- A combined measure that assesses the P/R tradeoff is F measure (weighted harmonic mean):

$$F = \frac{1}{a \frac{1}{P} + (1-a) \frac{1}{R}} = \frac{(b^2 + 1)PR}{b^2 P + R}$$

- People usually use balanced F_1 measure

- i.e., with $\beta = 1$ (that is, $\alpha = \frac{1}{2}$), then

$$F_1 = 2PR/(P+R)$$

$$F - measure = \frac{2 * Recall * Precision}{Recall + Precision}$$

Quiz

$$F_1 = 2PR/(P+R)$$

- $P = 40\%$ $R = 40\%$ $F_1 =$

- $P = 75\%$ $R = 25\%$ $F_1 =$

Quiz

$$F_1 = 2PR/(P+R)$$

- $P = 40\%$ $R = 40\%$ $F_1 = 40.0\%$

- $P = 75\%$ $R = 25\%$ $F_1 = 37.5\%$